GLOBAL TERRORISM INTELLIGENCE SYSTEM (GTIS)

A Multi-Modal Machine Learning & Spatiotemporal Risk Analytics Suite

PROJECT REPORT

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ABSTRACT

The Global Terrorism Intelligence System (GTIS) is an end-to-end, multi-modal artificial intelligence and machine learning platform engineered to extract predictive security intelligence, solve perpetrator attribution gaps, estimate casualty lethality bounds, and forecast regional threat volumes from the Global Terrorism Database (GTD). Encompassing 181,691 historical terrorist incidents recorded across 47 years (1970–2017), GTIS addresses critical intelligence limitations inherent in traditional counter-terrorism analysis.

GTIS operationalizes four specialized machine learning engines built primarily on optimized LightGBM gradient boosting architectures: (1) Perpetrator Group Attribution, which attributes responsibility for unclaimed attacks with 92.71% overall accuracy and 99.96% Top-3 accuracy; (2) Attack Success Prediction, utilizing cost-sensitive weight balancing to achieve a 0.9059 ROC-AUC and 0.9810 PR-AUC; (3) Casualty and Lethality Risk Estimation, integrating ordinal severity classification with 50th and 90th percentile quantile regressors to bound extreme mass-casualty risks; and (4) Spatiotemporal Hotspot Discovery and Regional Volume Forecasting, combining unsupervised DBSCAN density clustering (118 global threat hotspots) with LightGBM lag regressors achieving a test set R^2 of 0.7891.

To make these predictive models accessible to intelligence officers, defense analysts, and strategic decision-makers, GTIS includes a full-featured interactive web dashboard developed using Streamlit, Plotly, and Pydeck. The application provides executive summary metrics, interactive 3D geospatial maps, a live ML scenario prediction studio, and model diagnostic panels.

CHAPTER 1: INTRODUCTION AND PROJECT MOTIVATION

1.1 Background & Problem Statement

Global terrorism remains one of the most volatile and complex threats to national security, public safety, and economic stability worldwide. Over the past five decades, terrorist tactics have evolved from localized regional insurgencies to highly networked, multi-domain operations. Analyzing patterns across tens of thousands of incidents presents enormous analytical challenges. Defense and intelligence organizations are tasked with processing massive streams of historical and contemporary event data to discern tactical shifts, attribute responsibility for unclaimed attacks, evaluate the effectiveness of protective infrastructure, and allocate defensive resources to high-risk geographic regions.

Historically, terror incident analysis relied primarily on manual post-event investigation, qualitative threat matrix modeling, and retrospective statistical summaries. While valuable, these manual approaches suffer from three fundamental limitations: first, they are highly labor-intensive and slow; second, they struggle to model complex non-linear interactions across spatial, temporal, target, and weapon dimensions; and third, they fail to provide automated probabilistic forecasting for emerging operational scenarios.

1.2 Machine Learning in Counter-Terrorism Intelligence

Machine Learning (ML) and Data Science offer transformative capabilities for counter-terrorism analytics. By algorithmically processing large-scale historical databases such as the Global Terrorism Database (GTD), ML models can discover subtle behavioral signatures, quantify operational risk factors, and project multi-step regional threat trajectories. However, applying ML to national security data requires strict methodological rigor. Machine learning models trained on temporal event data are highly susceptible to data leakage if future information is inadvertently exposed during feature engineering or cross-validation. Furthermore, extreme class imbalance (e.g., high attack success rates, rare mass-casualty events) requires specialized loss functions and cost-sensitive training strategies.

1.3 Objectives & Project Scope

The primary objective of the Global Terrorism Intelligence System (GTIS) project is to build a robust, production-ready machine learning framework and web-based analytical environment. Specifically, the project encompasses five core goals:

1. Rigorous Data Cleaning & Feature Engineering: Preprocess the complete GTD dataset (181,691 records), impute missing spatial coordinates, encode categorical variables, and engineer temporal lag features.

2. Leakage-Free Temporal Validation: Enforce a strict chronological train/test split (1970–2012 training set vs. 2013–2017 unseen evaluation holdout set) to evaluate model performance realistically.

3. Multi-Model ML Architecture Development: Design, train, and validate four specialized model pipelines targeting Perpetrator Attribution, Attack Success Prediction, Casualty Risk Bounding, and Spatiotemporal Forecasting.

4. Unclaimed Attack Resolution: Apply the trained Attribution Model to solve historical unclaimed terror attacks, quantifying group likelihoods for intelligence analysis.

5. Interactive Web Intelligence Dashboard: Develop a modern, glassmorphic Streamlit web application providing 3D map visualizer, live ML simulation studio, and interactive diagnostic charts.

CHAPTER 2: DATASET ARCHITECTURE AND EXPLORATORY DATA ANALYSIS

2.1 Overview of the Global Terrorism Database (GTD)

The Global Terrorism Database (GTD) is an open-source database maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism (START) at the University of Maryland. GTD contains systematically collected data on domestic and international terrorist incidents from 1970 through 2017. The database records 181,691 unique incidents across 135 structured attributes, making it the most comprehensive unclassified terrorism database in existence.

2.2 Dataset Attributes and Taxonomy

GTD attributes are organized into seven major functional categories:

• Incident Identifiers & Timing: Unique GTD ID, Year (iyear), Month (imonth), Day (iday), Approximate Date, Extended Duration Indicator.

• Geographic Location: Region (region, region_txt), Country (country, country_txt), State/Province (provstate), City (city), Latitude (latitude), Longitude (longitude), Vicinity Indicator.

• Attack Characteristics: Primary Attack Type (attacktype1, attacktype1_txt), Secondary/Tertiary Attack Types, Success Indicator (success), Suicide Indicator (suicide), Claim of Responsibility (claimed).

• Weaponry Details: Primary Weapon Type (weaptype1, weaptype1_txt), Weapon Sub-types, Specific Explosive/Firearm Details.

• Target Taxonomy: Target Category (targtype1, targtype1_txt), Target Sub-type, Specific Target/Victim Entity, Nationality of Target.

• Perpetrator Group Data: Group Name (gname), Sub-group Details, Number of Perpetrators (nperps), Claim Characteristics.

• Casualty & Damage Metrics: Number of Fatalities (nkill), Number of Injured (nwound), Total Casualties (total_casualty), Property Damage Indicator (property), Extent of Property Damage.

2.3 Temporal Dynamics & Historical Waves (1970–2017)

Exploratory analysis of GTD temporal distributions reveals distinct historical waves of global terrorist activity. During the 1970s and 1980s, incident volume averaged between 1,000 and 5,000 events per year, primarily concentrated in Latin America and Western Europe driven by Cold War political insurgencies. Following a temporary decline in the late 1990s, global terrorism escalated dramatically after 2004, driven by conflicts in Iraq, Afghanistan, Syria, Nigeria, and Somalia. Peak annual volume occurred in 2014, with over 16,800 recorded incidents in a single year, before beginning a gradual decline through 2017.

2.4 Geographic Distribution & Threat Clusters

Geographically, terrorism is highly concentrated. Middle East & North Africa (MENA) and South Asia account for over 50% of all recorded incidents in GTD history. Sub-Saharan Africa, Southeast Asia, and South America represent secondary regional nodes. High-density tactical zones include the Tigris-Euphrates River Valley (Iraq/Syria), the Khyber Pass / FATA region (Pakistan/Afghanistan), the Lake Chad Basin (Nigeria/Cameroon), and the Mindanao archipelago (Philippines).

2.5 Casualty Distributions & Heavy-Tailed Outlier Analysis

Analysis of GTD casualty figures reveals extreme right-skewness. Over 45% of recorded incidents resulted in zero casualties, while approximately 35% involved between 1 and 4 casualties. However, the top 1% of incidents—such as the September 11 attacks, the 2014 Sinjar massacre, and major coordinated truck bombings—resulted in hundreds or thousands of fatalities. This extreme power-law distribution invalidates ordinary linear regression models relying on Gaussian error assumptions, motivating GTIS's ordinal tier classification and quantile regression framework.

CHAPTER 3: DATA PREPROCESSING, FEATURE ENGINEERING AND LEAKAGE CONTROL

3.1 Spatial Imputation & Coordinate Cleaning

Approximately 8.2% of raw GTD records lacked explicit latitude and longitude coordinates. Rather than discarding these records—which would introduce geographical sample bias—GTIS implemented a hierarchical spatial imputation pipeline. Missing coordinates were imputed using national centroid coordinates calculated from valid spatial records within the same country and province. Out-of-bounds geographic anomalies were validated and standardized.

3.2 Categorical Ordinal Encoding & High-Cardinality Management

GTD contains categorical variables with varying cardinality, ranging from 12 world regions to over 200 countries and thousands of perpetrator group strings. To optimize compatibility with tree-based LightGBM split algorithms, categorical features were mapped to numeric ordinal indices while maintaining explicit label lookup dictionaries. High-cardinality perpetrator names were standardized, grouping rare splinter factions under parent organizational umbrellas (e.g., aligning regional Boko Haram factions).

3.3 Feature Engineering Strategy

To capture complex domain relationships, GTIS constructed several composite feature groups:

1. Tactical Modus Operandi Vectors: Combined binary flags for suicide indicator, extended duration (>24 hrs), property damage, hostage taking, and responsibility claims into dense vector representations.

2. Temporal Lag Features: For time-series forecasting, computed multi-period historical lags (lag_1, lag_2, lag_3, lag_6, lag_12 months) and moving window statistics (3-month and 6-month rolling averages).

3. Ordinal Casualty Severity Tiers: Engineered a 5-tier ordinal target variable: Tier 0 (0 casualties), Tier 1 (1–4 casualties), Tier 2 (5–19 casualties), Tier 3 (20–99 casualties), Tier 4 (100+ mass casualties).

4. Known Group & Claim Indicators: Constructed binary features flagging whether an incident had an explicit group claim or matched known organizational signatures.

3.4 Chronological Train/Test Partitioning & Leakage Prevention

A critical vulnerability in temporal machine learning is data leakage—where future information inadvertently contaminates model training via random cross-validation. To guarantee strict methodological integrity, GTIS enforces a strict temporal train/test split:

• Historical Training Set (1970–2012): 121,940 incidents utilized for model training, target encoding, and hyperparameter alignment.

• Out-of-Sample Evaluation Set (2013–2017): 59,751 unseen future incidents reserved exclusively for final performance benchmarking.

CHAPTER 4: MACHINE LEARNING SUITE IMPLEMENTATION AND EVALUATION

4.1 Perpetrator Group Attribution Model

Unclaimed terrorist attacks represent over 45% of total GTD incidents. The Attribution Model formulates perpetrator identification as a multi-class classification problem over the top 15 most active global perpetrator organizations (Taliban, ISIL, Al-Shabaab, New People's Army, Maoists, Boko Haram, PKK, FARC, SL, CPI-Maoist, Hamas, Hizbul Mujahideen, etc.).

Architecture & Hyperparameters: Trained using a Multi-Class LightGBM Classifier (objective='multiclass', num_class=15, n_estimators=300, learning_rate=0.05, num_leaves=63, subsample=0.8).

Evaluation Results: Tested on the 2013–2017 evaluation holdout set, the Attribution Model achieved an overall classification accuracy of 92.71%, a Weighted F1-Score of 0.9269, and a Top-3 Categorical Accuracy of 99.96%. When applied to the 28,330 unclaimed attacks in the evaluation period, the model attributed 2,891 attacks to the Taliban, 603 to the NPA, 508 to Al-Shabaab, and 368 to Maoist insurgents.

4.2 Attack Success Prediction Model

Predicting whether an attempted attack will succeed allows security agencies to identify protective tactical factors and defense vulnerabilities. Because historical GTD attacks exhibit a high success rate (~88%), standard accuracy metrics are misleading. GTIS deployed a cost-sensitive LightGBM binary classifier utilizing positive weight adjustment (scale_pos_weight = 0.134).

Evaluation Results: On out-of-sample evaluation data, the model achieved an Area Under the ROC Curve (ROC-AUC) of 0.9059, a Precision-Recall AUC (PR-AUC) of 0.9810, and an overall classification accuracy of 85.34%, effectively isolating high-risk attack vectors from unsuccessful operational attempts.

4.3 Casualty & Lethality Risk Estimation System

To address extreme casualty right-skewness, GTIS deploys a dual-component casualty prediction system:

1. Ordinal Severity Classifier: LightGBM multi-class classifier predicting discrete casualty tiers (Tier 0 to 4), achieving a Macro F1 of 0.4905 and Weighted F1 of 0.7029.

2. Quantile Regressors: Two LightGBM regressors trained with Pinball Loss (quantile loss). The median regressor (alpha = 0.50) estimates expected baseline casualties (loss = 2.21), while the upper-bound regressor (alpha = 0.90) estimates high-risk 90th percentile lethality (loss = 2.20).

4.4 Spatiotemporal Hotspot Discovery & Regional Volume Forecasting

GTIS combines unsupervised spatial density analysis with supervised time-series forecasting:

• Unsupervised Spatial Density Clustering: DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identified 118 distinct global threat hotspots using spatial parameters (eps = 1.5°, min_samples = 50). Primary hotspots include Baghdad (Iraq), FATA (Pakistan), Borno (Nigeria), and Helmand (Afghanistan).

• Time-Series Volume Regressor: LightGBM regressor trained on regional monthly lag features achieved an out-of-sample evaluation R^2 of 0.7891 and a Mean Absolute Error (MAE) of 33.81 incidents/month across all world regions.

CHAPTER 5: INTERACTIVE WEB DASHBOARD SYSTEM

5.1 System Architecture & Framework Selection

To make the machine learning suite actionable for intelligence officers, defense analysts, and strategic researchers, GTIS features an interactive web application (app.py) developed using Streamlit, Plotly Express, Pydeck, and Joblib. The application operates natively in Python, loading serialized model binaries (models/*.joblib) and pre-calculated metrics (outputs/metrics.json).

5.2 Tab 1: Executive Summary & Global GTD Overview

Tab 1 presents a strategic macro-view of global terrorism dynamics. Top KPI cards display total recorded incidents (181,691), attribution model accuracy (92.7%), success ROC-AUC (0.906), spatial hotspots (118), and solved attribution gap (45.6%). Interactive Plotly charts display regional incident counts, top attack methods, target type breakdowns, and a 47-year temporal incident volume line chart.

5.3 Tab 2: 3D Geospatial Threat Hotspots

Tab 2 provides interactive geospatial exploration of terror incidents and DBSCAN threat clusters. Built on Pydeck and Mapbox carto-darkmatter tiles, users can filter map displays by world region, country, and year range (1970–2017). Hover tooltips reveal incident year, attack type, perpetrator group, and total casualties, while a side panel details top spatial hotspot centroids.

5.4 Tab 3: Live ML Model Prediction Studio

Tab 3 enables interactive scenario simulation. Analysts configure tactical incident parameters - including Incident Year, Month, Region, Country, Attack Type, Target Category, Weapon Type, Suicide Indicator, Extended Duration, Property Damage, and Responsibility Claim; and click 'Run Live ML Inference Engine'. The system computes instant predictions across all 4 models:

• Perpetrator Attribution Card: Displays top predicted group and confidence bar chart.

• Attack Success Card: Displays predicted outcome and gauge chart showing success probability.

• Casualty Risk Card: Displays predicted severity tier, median casualty estimate (q=0.50), 90th percentile high-risk upper bound (q=0.90), and tier probability chart.

5.5 Tab 4: Forecasting & Model Diagnostics

Tab 4 presents model evaluation benchmarks and feature importance rankings. Users can select any of the four ML models to inspect top 15 feature importances, evaluating the relative influence of geographic coordinates, target types, and weapon selections.

CHAPTER 6: EXPERIMENTAL RESULTS AND BENCHMARKS

Table 6.1 consolidates quantitative evaluation metrics across all four GTIS machine learning modules evaluated on the unseen 2013–2017 holdout evaluation dataset.

Model Domain	Algorithm Architecture	Primary Evaluation Metric	Secondary Evaluation Metric
Perpetrator Attribution	Multi-Class LightGBM	Accuracy: 92.71%	Top-3 Acc: 99.96%
Attack Success Prediction	Cost-Sensitive LightGBM	ROC-AUC: 0.9059	PR-AUC: 0.9810
Casualty Risk Estimation	Quantile LightGBM (q=0.50, 0.90)	Pinball Loss (0.50): 2.21	Pinball Loss (0.90): 2.20
Spatiotemporal Forecasting	Regional Lag LightGBM	Test R^2: 0.7891	Test MAE: 33.81

Table 6.1

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Summary of Contributions

The Global Terrorism Intelligence System (GTIS) establishes a complete, leakage-free machine learning and predictive intelligence framework for the Global Terrorism Database. Key technical accomplishments include:

1. High-Accuracy Perpetrator Attribution: Solved over 28,000 unclaimed terror attacks with 92.71% accuracy and 99.96% Top-3 accuracy.

2. Robust Success & Casualty Risk Models: Isolated tactical success factors (ROC-AUC 0.9059) and bounded extreme casualty distributions using quantile regression.

3. Spatiotemporal Hotspot Discovery: Discovered 118 spatial threat clusters and achieved high-precision regional volume forecasting (R^2 0.7891).

4. Operational Web Dashboard: Built an interactive, dark-themed Streamlit application enabling live scenario simulation and 3D geospatial mapping.

7.2 Future Enhancements

Future research directions for GTIS include processing unstructured GTD text summaries using Sentence-BERT (SBERT) embeddings, incorporating real-time streaming news API ingestion, and deploying the web suite to cloud infrastructure for collaborative counter-terrorism analytics.

REFERENCES

1. National Consortium for the Study of Terrorism and Responses to Terrorism (START). (2018). Global Terrorism Database [Data file]. Retrieved from https://www.start.umd.edu/gtd

2. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems, 30, 3146-3154.

3. Ester, M., Kriegel, H. P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In KDD-96 (Vol. 96, No. 34, pp. 226-231).

4. Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.

5. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.